

Biomedical Entity Extraction and Patient-Friendly Explanation from Clinical Reports Using BioBERT and MedQuAD

Adithya Vardhan Selvam
University of Florida
ad.selvam@ufl.edu

Abstract—Clinical reports often contain important information about diseases, medications, symptoms, and observations, but much of this information is written in unstructured free-text form. This makes it difficult for patients, students, and downstream healthcare applications to quickly identify clinically meaningful terms and understand their relevance. This paper presents MedAgent, an end-to-end biomedical natural language processing system that extracts disease and chemical entities from patient reports and provides patient-friendly explanations for the extracted terms. The system fine-tunes BioBERT on the BC5CDR biomedical named entity recognition dataset and integrates MedQuAD-based retrieval to connect extracted entities with medical question-answer explanations. Synthetic patient reports generated using Synthea and manually prepared sample reports are used to test the complete workflow. The system is deployed through a Gradio interface that supports both text-file upload and direct text entry. Experimental evaluation on the BC5CDR benchmark demonstrates strong entity recognition performance, with a precision of 0.8416, recall of 0.8919, F1-score of 0.8660, and accuracy of 0.9665. Qualitative demonstrations show that the system can identify diseases such as asthma, hypertension, prediabetes, and coronary artery disease, along with medications such as Albuterol, Prednisone, Aspirin, and Atorvastatin. This work contributes a complete AI lifecycle pipeline that combines biomedical entity extraction, explanation retrieval, interface design, and responsible AI considerations for interpretable medical text understanding.

Index Terms—Biomedical Natural Language Processing, BioBERT, Named Entity Recognition, MedQuAD, Clinical Text Mining, Explainable AI

I. INTRODUCTION

Clinical documentation contains a large amount of valuable medical information. Patient reports, physician notes, discharge summaries, and synthetic health records often describe diseases, medications, symptoms, laboratory observations, and treatment plans in free-text form. Although this format is useful for human communication, it creates challenges for computational analysis. Important medical terms may be buried inside long paragraphs, written with varied phrasing, or mixed with non-clinical information. As a result, unstructured reports are difficult to search, summarize, or convert into patient-friendly explanations without automated text processing.

The problem addressed in this work is the lack of a simple end-to-end system that can take a patient-style medical report, identify important biomedical entities, and explain them in a readable way. Many biomedical NLP systems focus only on

entity extraction or only on question answering. In contrast, this work combines entity recognition with explanation retrieval and presents the results through an interactive interface. The resulting system is intended as a prototype for biomedical report understanding, not as a diagnostic or treatment decision tool.

The objective of this work is to design and implement MedAgent, a biomedical entity extraction and explanation system. The system accepts patient reports as text input, uses a fine-tuned BioBERT model to identify biomedical entities, groups the extracted terms into disease and chemical categories, retrieves relevant explanations from MedQuAD, and displays the output in a structured Gradio interface. The main goal is to make clinical text easier to inspect and understand by converting raw text into categorized entities and readable explanations.

This work covers the full AI lifecycle, including dataset selection, preprocessing, model fine-tuning, evaluation, inference, user interface design, qualitative testing, and responsible AI reflection. The system uses BC5CDR for supervised named entity recognition training, MedQuAD for explanation retrieval, and Synthea-style patient reports for realistic testing. The final prototype demonstrates a complete workflow from raw patient report input to structured and interpretable output.

The main contributions of this work are as follows:

- 1) A fine-tuned BioBERT model for biomedical named entity recognition over disease and chemical entities.
- 2) A complete inference pipeline that reconstructs entity spans from token-level BIO predictions.
- 3) A MedQuAD-based explanation module that links extracted entities to patient-friendly question-answer content.
- 4) A fallback explanation layer for common medications and clinical terms not directly covered by MedQuAD.
- 5) A Gradio-based user interface that supports file upload, text entry, detailed entity tables, grouped output, and summary generation.
- 6) An evaluation of model performance using precision, recall, F1-score, accuracy, and confusion matrix analysis.
- 7) A responsible AI analysis addressing limitations, dataset bias, explanation reliability, and non-diagnostic use.

The rest of this report is organized as follows. Section II reviews related work in biomedical NLP, BioBERT, BC5CDR,

MedQuAD, synthetic patient data, and responsible AI. Section III describes the system design and implementation. Section IV explains the AI lifecycle stages, including data collection, preprocessing, model training, and interface design. Section V presents evaluation results. Section VI discusses strengths, limitations, and work novelty. Section VII describes future work. Section VIII presents responsible AI considerations. Section IX concludes the report.

II. RELATED WORK

Biomedical natural language processing has become an important research area because large amounts of healthcare and biomedical knowledge exist in text form. Clinical notes, PubMed abstracts, patient education resources, and medical question-answer datasets all contain information that can support information extraction and decision support systems. However, biomedical language differs from general-domain language because it contains specialized terminology, abbreviations, chemical names, disease names, and domain-specific relationships. This creates challenges for general language models.

BERT introduced deep bidirectional transformer representations and showed that a pre-trained model could be fine-tuned for many NLP tasks with limited task-specific architecture changes [1]. However, applying general-domain BERT directly to biomedical text can be limited because biomedical documents contain vocabulary and phrasing patterns that differ from ordinary web or book text. BioBERT addresses this limitation by continuing BERT pretraining on biomedical corpora such as PubMed abstracts and PMC full-text articles [2]. BioBERT has shown strong performance on biomedical named entity recognition, relation extraction, and question-answering tasks. This work builds on BioBERT because the task requires understanding domain-specific medical terms such as “Acetaminophen,” “coronary artery disease,” “hypertension,” and “shortness of breath.”

Named entity recognition is a common task in biomedical text mining. The goal is to locate spans of text corresponding to meaningful biomedical concepts. The BC5CDR corpus is widely used for chemical and disease recognition and chemical-disease relation extraction [3]. It provides annotated mentions of chemical and disease entities in PubMed articles. This makes it suitable for training a biomedical NER model that can distinguish disease entities from chemical entities. In this work, BC5CDR is used for supervised fine-tuning of BioBERT as a token classification model.

Medical question-answering resources are also important for interpretability. MedQuAD is a medical question-answer dataset created from trusted medical resources, including NIH sources [4]. It contains medical questions and corresponding answers covering diseases, treatments, symptoms, diagnosis, and other healthcare topics. This system uses MedQuAD not as a generative model training dataset, but as a retrieval resource. After entities are extracted from a report, the system attempts to retrieve relevant question-answer pairs. This allows the system to go beyond extraction by offering explanations.

Synthetic patient data has become useful for healthcare AI prototyping because real patient data is sensitive and difficult to share. Synthea is an open-source synthetic patient generator that creates realistic electronic health record data while avoiding direct use of real patient identities [5]. In this work, Synthea-style reports and manually prepared sample reports are used to test the user-facing pipeline. Synthetic reports allow the system to be demonstrated without exposing private patient information.

Transformer libraries and rapid prototyping tools also support the development of practical AI systems. The Hugging Face Transformers library provides a unified API for loading, fine-tuning, and evaluating transformer models [6]. The implementation uses Hugging Face Transformers and PyTorch for the BioBERT implementation. Gradio is used for the user interface because it supports fast deployment of machine learning demos and allows users to interact with models through file upload and text input [7].

Existing biomedical NLP systems often focus on a single component, such as entity extraction, relation extraction, or question answering. The novelty of this work is not the creation of a new transformer architecture, but the integration of multiple components into a complete and usable pipeline. MedAgent combines biomedical NER, explanation retrieval, fallback explanations, grouped output, and a user-facing interface. This makes the system more practical for interactive biomedical text analysis than a standalone model evaluation script.

Responsible AI is especially important in medical AI systems. The World Health Organization emphasizes that AI for health should be transparent, safe, accountable, and aligned with human well-being [8]. This work follows that principle by limiting the system’s role to information extraction and explanation. It does not generate diagnoses, recommend treatments, or replace clinical judgment. The interface and report clarify that outputs are informational and may require professional review.

III. SYSTEM DESIGN AND IMPLEMENTATION

A. System Overview

MedAgent is designed as a modular biomedical text processing pipeline. The system accepts a patient report, preprocesses the text, applies a BioBERT-based named entity recognition model, reconstructs entity phrases, groups the entities into categories, retrieves explanations, and displays the final output through a Gradio interface.

The architecture contains six major components:

- 1) Input Module: Accepts uploaded .txt patient reports or pasted medical text.
- 2) Preprocessing Module: Standardizes input text and prepares it for tokenization.
- 3) BioBERT NER Module: Performs token-level classification using a fine-tuned BioBERT model.
- 4) Entity Reconstruction Module: Converts token predictions into complete disease and chemical phrases.

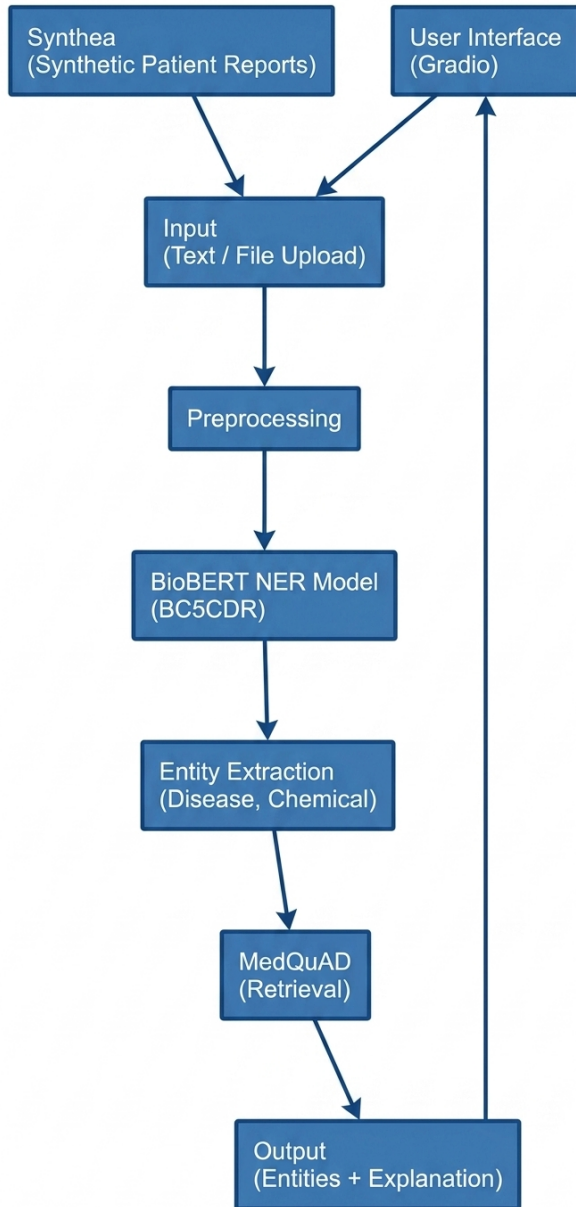


Fig. 1. System architecture of MedAgent showing the flow from patient report input to preprocessing, BioBERT NER, entity extraction, MedQuAD retrieval, fallback explanation, grouped output, and Gradio interface display.

- 5) Explanation Retrieval Module: Searches MedQuAD and fallback dictionaries for explanations.
- 6) Interface Module: Displays processed text, detailed entity table, grouped output, and summary.

The system is intentionally modular. Each component can be improved independently. For example, the keyword-based MedQuAD retrieval method can later be replaced with embedding-based semantic retrieval without changing the BioBERT model. Similarly, the Gradio interface can be redesigned without changing the NER training logic.

B. Computing Architecture

The system was implemented in Python using PyTorch, Hugging Face Transformers, Hugging Face Datasets, pandas, and Gradio. Persistent storage was used to save the trained BioBERT model, tokenizer files, processed datasets, and result artifacts, enabling reproducible inference without retraining.

This design is suitable for a lightweight biomedical NLP prototype because it separates model training, persistent storage, inference, retrieval, and interface execution. Training a transformer model can be expensive, but once the model is fine-tuned and saved, inference can be performed by loading the model from disk. During inference, the system loads the saved model checkpoint rather than retraining the model. It loads the saved model and tokenizer directly from the `biobert_bc5cdr_model` folder, then loads the saved MedQuAD dataset from the system directory.

C. Data Sources

The system uses three main data sources.

BC5CDR: This dataset is used for supervised NER training. It contains disease and chemical annotations and is suitable for token classification. The model learns to assign BIO labels such as B-Disease, I-Disease, B-Chemical, I-Chemical, and O.

MedQuAD: This dataset is used for explanation retrieval. For each extracted entity, the system searches MedQuAD question, question focus, and synonym fields. If a matching question-answer pair is found, the answer is displayed as the explanation.

Synthea and sample patient reports: These are used for qualitative testing. Synthetic reports help simulate realistic patient-style inputs without exposing real patient data. In the qualitative evaluation, sample reports include respiratory disease, heart disease, and prediabetes/anemia examples.

D. Model Selection

BioBERT was selected because it is a domain-specific transformer model designed for biomedical language. General BERT models may understand grammar and general semantics, but biomedical terms often require domain-specific representations. BioBERT improves this by using biomedical corpora during pretraining.

The model is used as a token classification model. A classification head is added on top of BioBERT to predict the entity label for each token. The output is then post-processed to reconstruct entity phrases. This is appropriate for NER because the task requires identifying both the entity type and the boundaries of the entity span.

E. Training Configuration

The training configuration used in this work is summarized in Table I.

The model is trained using the Hugging Face Trainer API. The Trainer simplifies batching, validation, logging, checkpointing, and metric computation. The best-performing checkpoint is selected based on validation F1-score and saved for inference.

TABLE I
TRAINING CONFIGURATION FOR BIOBERT NER

Component	Configuration
Base model	dmis-lab/biobert-base-cased-v1.2
Task	Biomedical named entity recognition
Dataset	BCSCDR
Entity types	Disease, Chemical
Labeling scheme	BIO tagging
Epochs	3
Learning rate	2×10^{-5}
Batch size	8
Optimizer	AdamW
Framework	Hugging Face Transformers + PyTorch
Model selection	Best checkpoint based on F1-score

F. Tokenization and Label Alignment

A key implementation challenge in transformer-based NER is token-label alignment. BioBERT uses WordPiece tokenization, which means that a single biomedical word may be split into multiple subword tokens. For example, a term such as “Azithromycin” or “hyperlipidemia” may be divided into smaller token pieces depending on the tokenizer vocabulary.

Because the dataset labels are usually aligned at the word or span level, the labels must be mapped correctly to the tokenizer output. If alignment is incorrect, the model may learn wrong entity boundaries. The implementation handles this by assigning labels to the correct token positions and ensuring continuation subwords are treated consistently. During inference, subword tokens are merged back into readable entity phrases.

G. Entity Reconstruction

After model inference, the model outputs token-level labels. These labels are not directly user-friendly because they are assigned to individual tokens. The entity reconstruction module converts token predictions into complete entity spans. It detects B- labels as the start of entities and combines following I- labels of the same type. It also removes special tokens and merges WordPiece fragments.

For example, if the model predicts that “shortness,” “of,” and “breath” are part of the same disease phrase, the reconstruction module outputs “shortness of breath” as one entity. This is important for usability because users expect full medical phrases rather than isolated tokens.

H. Explanation Retrieval

The explanation module uses a two-stage strategy.

First, it searches the MedQuAD lookup table. The lookup table stores question, question focus, synonyms, answer, and lowercase versions of searchable fields. The retrieval function first checks for strong exact matches against the question focus. If no exact match exists, it checks whether the entity appears in the question or synonyms. Rows with missing or None answers are skipped.

Second, the system uses fallback explanations for common clinical terms and medications. This coverage layer is included because MedQuAD does not always contain direct explanation

rows for medication names such as Albuterol, Prednisone, Azithromycin, Acetaminophen, Aspirin, Atorvastatin, Metformin, and Lisinopril. The fallback dictionary provides simple explanations for common entities so that the interface does not display empty outputs for important terms.

This hybrid strategy improves explanation coverage and output completeness. MedQuAD provides dataset-grounded explanations when available, while fallback explanations improve coverage for common medicines and symptoms.

The retrieval component also improves the interpretability of the overall system by connecting model predictions to explanatory context. Instead of only displaying extracted terms, the system provides a short explanation that helps users understand why each term may be clinically meaningful. This is especially important for medication names and multi-word disease expressions, where the entity may be correctly extracted but not immediately understandable to a non-specialist user.

I. Human-Computer Interaction Design

The system uses a Gradio interface to make the model interactive. The interface supports two input modes: uploading a patient report as a .txt file or pasting medical text directly into a textbox. This flexibility allows both realistic report-based testing and quick manual testing.

The interface provides four output sections:

- 1) Processed Input Text: Displays the report that was analyzed.
- 2) Extracted Entities and Explanations: Shows each entity, entity type, matched question, and explanation.
- 3) Grouped Biomedical Entities: Separates diseases from chemicals.
- 4) Summary: Provides entity counts and category-level overview.

This layout separates raw input, model predictions, retrieved explanations, and summarized outputs into distinct regions. Separating these outputs reduces cognitive load and allows users to verify how each extracted entity is connected to the original report. The interface also supports repeated testing with different reports, which was useful for validating the consistency of the inference and retrieval pipeline.

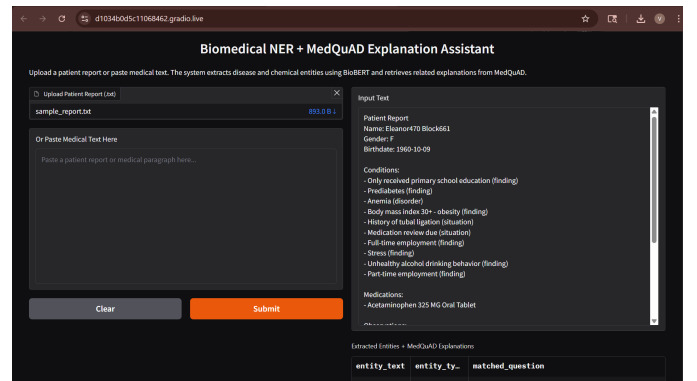


Fig. 2. MedAgent v2 interface showing upload, text input, Analyze button, Clear button, and processed patient report display.

TABLE II
BIOBERT EVALUATION RESULTS ON BC5CDR

Metric	Score
Precision	0.8416
Recall	0.8919
F1-score	0.8660
Accuracy	0.9665

The F1-score of 0.8660 indicates strong overall performance for biomedical entity recognition. This is the most important metric because it balances false positives and false negatives. The recall score of 0.8919 is especially valuable because biomedical systems should avoid missing important disease or medication mentions. The precision score of 0.8416 shows that most predicted entities are correct, although some false positives remain.

The accuracy score of 0.9665 is high, but it should be interpreted carefully. In token classification, most tokens are not entities and belong to the O class. Therefore, accuracy can appear high even when entity boundary detection is imperfect. Precision, recall, and F1-score are more meaningful for evaluating NER quality.

In the context of this project, recall is especially important because the system is designed to help users discover relevant biomedical terms from patient reports. If the model misses a disease or medication mention, the explanation module will never receive that entity and the user may not see an important part of the report. Therefore, the recall score of 0.8919 is a strong result for the intended use case. It shows that the model is effective at identifying most relevant biomedical mentions in the test data.

Precision is also important because incorrect entity predictions can reduce user trust. A precision score of 0.8416 means that most extracted entities are valid, but the system still produces some false positives. In practice, these false positives may appear as extra symptoms, partial disease phrases, or terms that are medically related but not the main condition being described. This behavior was also observed during qualitative testing, where the model sometimes extracted symptom-like phrases as disease entities. However, for an explanatory prototype, a slightly higher recall is useful because it allows the user to inspect more candidate medical terms rather than missing them completely.

The F1-score of 0.8660 provides the best single summary of model performance because it balances precision and recall. This indicates that the fine-tuned BioBERT model learned the biomedical entity recognition task effectively. Since the model was trained on BC5CDR and then tested through patient-style reports, the result also suggests that domain-specific pretraining helps the model recognize biomedical terminology beyond the exact training examples.

The evaluation results also show that the model is suitable for the final inference workflow. Since the model does not need to be retrained during inference, the saved checkpoint can be loaded directly and used to process new reports. This supports the practical goal of the project: building a

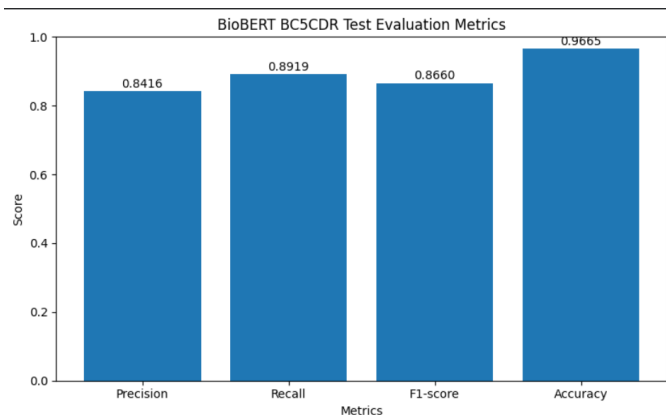


Fig. 4. Bar chart showing BioBERT evaluation metrics: precision, recall, F1-score, and accuracy.

usable biomedical report understanding system rather than only reporting benchmark scores.

B. Confusion Matrix Interpretation

The confusion matrix shows that most predictions lie on the diagonal, meaning that the model correctly classifies most tokens. The strongest performance is observed for the non-entity class and chemical labels. Disease labels also perform well but show slightly more boundary-related errors.

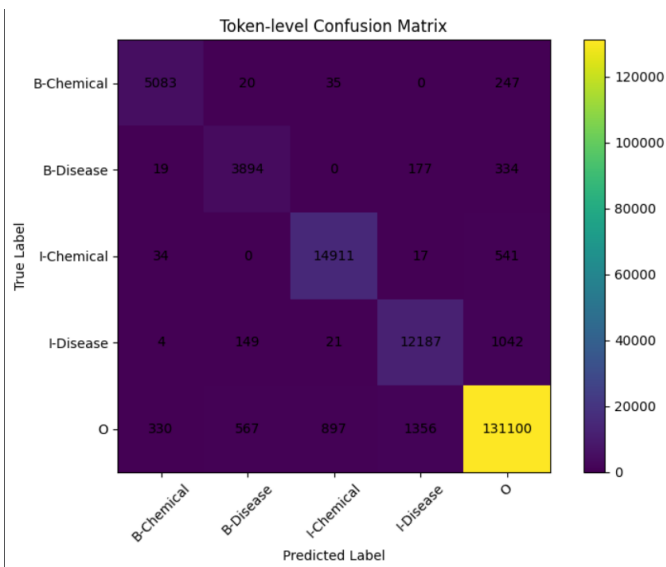


Fig. 5. Token-level confusion matrix showing classification behavior across B-Chemical, I-Chemical, B-Disease, I-Disease, and O labels.

Boundary errors are expected in sequence labeling. The model must not only identify that a disease or chemical exists, but also determine the exact start and end of the phrase. Multi-token disease names such as “shortness of breath,” “coronary artery disease,” and “chronic kidney disease” are more challenging than single-token chemical names.

The confusion matrix also helps explain the difference between chemical and disease extraction behavior. Chemical entities tend to have clearer lexical patterns because many medication names and chemical names are distinctive. As a result, the model can often recognize them with fewer boundary errors. Disease entities are more complex because they may appear as short single-word terms, multi-word phrases, symptoms, or descriptive clinical expressions. Examples include “asthma,” “chest pain,” “shortness of breath,” and “coronary artery disease.” These phrases require the model to identify both the correct entity type and the complete span.

Most remaining errors are related to boundary detection rather than complete failure to recognize biomedical content. For example, the model may extract a phrase that is slightly shorter or longer than expected. This is a common issue in token classification models because the model predicts labels one token at a time. Even when the general medical concept is detected correctly, the exact start and end token may not always match the gold annotation.

For the system goal, these errors are manageable because the interface displays extracted entities in a transparent table. However, for a production clinical system, boundary errors would need to be reduced through additional training data, post-processing rules, confidence filtering, or entity normalization.

Overall, the confusion matrix supports the quantitative results by showing that the model has learned the main token classification patterns effectively. The remaining errors are concentrated around more difficult entity-boundary decisions, which is expected in biomedical NER because medical phrases often vary in length, structure, and context.

C. Qualitative Demonstration: Asthma Report

This case study was selected because it contains both symptom-like disease mentions and medication names, which allows the system to be evaluated on two important biomedical entity categories. Respiratory reports are also useful for qualitative testing because the same report may include clinical conditions, symptoms, suspected infections, and prescribed medications. In this example, the extracted entities show that the model can identify both single-word terms such as cough and fever and multi-word expressions such as shortness of breath and airway inflammation.

The asthma sample report contains respiratory symptoms and medications. The report includes terms such as cough, wheezing, shortness of breath, fever, asthma, acute bronchitis, Albuterol, Prednisone, Azithromycin, and Acetaminophen.

The system successfully extracts respiratory disease-related terms and medication-related terms. It groups cough, wheezing, shortness of breath, fever, asthma, airway inflammation, and bacterial bronchitis under diseases. It groups Albuterol, Prednisone, Azithromycin, and Acetaminophen under chemicals.

TABLE III
QUALITATIVE OUTPUT EXAMPLE FOR ASTHMA REPORT

Entity	Type	Explanation Summary
cough	Disease	Reflex that clears throat and airways
wheezing	Disease	High-pitched breathing sound from narrowed airways
shortness of breath	Disease	Difficulty breathing or getting enough air
fever	Disease	Increased body temperature often caused by infection
Asthma	Disease	Chronic breathing condition involving inflamed airways
Albuterol	Chemical	Bronchodilator used to improve breathing
Prednisone	Chemical	Corticosteroid used to reduce inflammation
Azithromycin	Chemical	Antibiotic used for bacterial infections
Acetaminophen	Chemical	Used to reduce fever and relieve pain

This result demonstrates that the pipeline can process a realistic patient-style report and return both entity categories and explanations. The fallback layer is especially useful here because several medication explanations are not reliably covered by direct MedQuAD matching.

D. Qualitative Demonstration: Heart Disease Report

The heart disease sample report includes coronary artery disease, hypertension, hyperlipidemia, chest pain, shortness of breath, Aspirin, Atorvastatin, Metoprolol, and Nitroglycerin. The system extracts both disease and chemical entities and provides explanations.

fatigue	Disease	What is fatigue?	Fatigue means unusual tiredness or lack of energy. It can be caused by illness, stress, infection, poor sleep, or
coronary artery disease	Disease	What is coronary artery disease?	Coronary artery disease occurs when the blood vessels supplying the heart become narrowed or blocked, often causing
hypertension	Disease	What is hypertension?	Hypertension, or high blood pressure, means the force of blood against artery walls is consistently too high.
hyperlipidemia	Disease	What is hyperlipidemia?	Hyperlipidemia means there are high levels of fats such as cholesterol or triglycerides in the blood.
shortness of breath	Disease	What are the signs and symptoms of congestive heart failure?	What are the signs and symptoms of congestive heart failure? The heart pumps blood to the rest of the body.
aspirin	Chemical	What is aspirin used for?	Aspirin is used to reduce pain, fever, and inflammation. It may also help reduce the risk of heart disease and stroke.
atorvastatin	Chemical	What is atorvastatin used for?	Atorvastatin is used to lower cholesterol levels and reduce the risk of heart disease and stroke.
metoprolol	Chemical	What is metoprolol used for?	Metoprolol is a beta blocker used to treat high blood pressure, chest pain, and some heart conditions.
nitroglycerin	Chemical	What is nitroglycerin used for?	Nitroglycerin is used to relieve chest pain by relaxing blood vessels and improving blood flow to the heart.
cholesterol	Chemical	Do you have information about cholesterol?	Summary : Cholesterol is a waxy, fat-like substance that occurs naturally in all parts of the body. Your body can

Fig. 6. MedAgent output for the heart disease sample report showing extracted cardiovascular disease entities, medication entities, grouped biomedical entities, and summary output.

Fig. 6 shows the final interface output for the heart disease case study. The system extracts disease entities such as chest pain, shortness of breath, coronary artery disease, hypertension, and hyperlipidemia, along with chemical entities such as Aspirin, Atorvastatin, Metoprolol, Nitroglycerin, and Cholesterol. The grouped output and summary make the result easier to interpret by separating clinical conditions from medications and chemical-related terms.

For example, coronary artery disease is explained as narrowing or blockage of blood vessels supplying the heart. Hypertension is explained as high blood pressure. Atorvastatin is explained as a medication used to lower cholesterol, while Nitroglycerin is explained as a medication used to relieve chest pain by improving blood flow.

This test demonstrates that MedAgent generalizes beyond respiratory reports and can process cardiovascular content. It also shows the value of grouped output because heart conditions and medications can be separated clearly.

E. Qualitative Demonstration: Original Patient Report

The original sample report includes prediabetes, anemia, obesity, stress, unhealthy alcohol behavior, Acetaminophen, Hemoglobin A1c, BMI, blood pressure, heart rate, respiratory rate, and glucose. The system extracts disease-related entities such as prediabetes, anemia, obesity, pain, and stress, along with chemical-related terms such as Acetaminophen and glucose.

This test is useful because it contains both clinical and non-clinical information. Some social findings such as education and employment are not directly relevant to biomedical NER. This highlights a limitation: patient reports may include many types of information, but the current model is trained mainly for disease and chemical extraction. Future versions could support additional categories such as procedures, lab tests, social determinants of health, and vital signs.

F. Interface Evaluation

The interface supports both detailed model inspection and concise output interpretation.

From a results perspective, the interface is important because it demonstrates that the model can be used in an interactive workflow. A high F1-score alone does not prove that the system is usable. The Gradio interface shows that the trained model, retrieval module, and output formatting can work together in real time. This supports the goal of building an integrated biomedical NLP system rather than only a model training experiment.

The interface outputs also make model behavior easier to analyze. When an entity is extracted incorrectly or an explanation is not available, the issue can be seen directly in the table. This helped guide the final refinement of the system, including the addition of fallback explanations and duplicate removal. Therefore, the interface served not only as a user-facing application but also as a debugging and evaluation tool during development.

VI. DISCUSSION

A. Strengths

The main strength of MedAgent is that it provides a complete end-to-end biomedical NLP pipeline. It does not stop at model training. It includes data preprocessing, model fine-tuning, evaluation, inference, explanation retrieval, interface design, and qualitative testing. This makes the system closer

to a usable AI application than a standalone notebook experiment.

Another strength is the use of a biomedical domain-specific model. BioBERT is better suited for clinical and biomedical terminology than a general language model. This is important because the input reports contain terms such as “Acetaminophen,” “hyperlipidemia,” “Hemoglobin A1c,” and “coronary artery disease.”

The system also improves interpretability by connecting extracted entities to explanations. Entity extraction alone tells the user what terms were found, but explanation retrieval helps the user understand what those terms mean. The grouped output further improves readability by separating diseases from chemicals.

The final system also demonstrates practical engineering improvements. The trained model is saved and reloaded instead of retrained during every run. The MedQuAD lookup table is prebuilt for faster retrieval. Fallback explanations improve output completeness. These changes improve inference stability, response consistency, and reproducibility.

A further strength is that the pipeline is modular. The entity extraction model, retrieval component, fallback explanation layer, and interface can be modified independently. This makes the system easier to extend in future work, especially if semantic retrieval, additional entity types, or improved explanation sources are added.

B. Limitations

The first limitation is retrieval quality. The current MedQuAD retrieval method is based mainly on lexical matching. This means that it may miss relevant explanations when the entity phrasing differs from the dataset phrasing. It may also retrieve loosely related explanations when a term appears in a broader question. Although fallback explanations help, they are manually defined and do not scale automatically.

The second limitation is entity coverage. BC5CDR focuses on diseases and chemicals. Patient reports often include many other important categories such as lab tests, procedures, vital signs, anatomical locations, symptoms, social history, and lifestyle factors. The current system may extract some symptoms as disease entities, but it is not trained as a full clinical concept extraction system.

The third limitation is input length. BioBERT has a maximum token length, commonly 512 tokens. Longer patient reports may require chunking and aggregation. Without chunking, important entities in later parts of long reports may be missed.

The fourth limitation is lack of clinical reasoning. MedAgent extracts and explains entities, but it does not diagnose patients, check medication interactions, assess severity, or generate treatment recommendations. This is intentional for safety reasons, but it limits the system’s use to report understanding and educational explanation.

Another limitation is that the current explanation layer does not assign confidence scores to retrieved answers. As a result, users can see the explanation text, but they cannot directly

measure how strongly the explanation matches the extracted entity. Future versions should include retrieval confidence, model confidence, or source labels to make the output more transparent.

C. Novelty and Contributions

The novelty of this work lies in the integration of several components into a complete interpretable biomedical report understanding system. The system combines:

- 1) BioBERT-based biomedical entity extraction.
- 2) BC5CDR supervised fine-tuning.
- 3) MedQuAD explanation retrieval.
- 4) Fallback explanations for common terms.
- 5) Grouped output by entity category.
- 6) Interactive Gradio interface.
- 7) End-to-end evaluation using sample patient reports.

This combination creates a practical prototype that is more interpretable and application-oriented than a model-only implementation. The system shows how entity extraction can be connected to explanation retrieval to support patient-friendly interpretation of medical text.

The contribution is therefore not limited to model accuracy alone. The work demonstrates how a biomedical NER model can be integrated with retrieval, post-processing, and interface design to form a complete information extraction workflow. This is important because real-world biomedical NLP systems require more than benchmark performance; they also require usable outputs, transparent presentation, and responsible communication of limitations.

D. Broader Impact

A system like MedAgent could support education, clinical NLP prototyping, patient report summarization, and healthcare data preprocessing. For educational settings, it illustrates how transformer models can be applied to biomedical text. For patients, a future improved version could help explain complex medical reports in simpler terms. For developers, it provides a modular architecture that can be extended with semantic search, clinical ontologies, or LLM-based summarization.

The system may also support early-stage clinical text analytics by helping transform unstructured reports into categorized biomedical entities. This could be useful in workflows where large collections of reports need to be inspected, summarized, or prepared for downstream analysis. However, such use would require stronger validation, privacy protection, and expert review before deployment in real healthcare environments.

Therefore, the system must be used carefully because biomedical outputs can be misinterpreted.

VII. FUTURE WORK AND IMPROVEMENTS

Several improvements can be made in future versions of MedAgent.

First, the keyword-based MedQuAD retrieval module should be replaced or enhanced with semantic retrieval. Embedding models such as BioSentVec, Sentence-BERT, or biomedical sentence transformers could be used to encode

entities and MedQuAD questions. This would allow the system to retrieve explanations based on meaning rather than exact word overlap.

Second, the system should support more biomedical entity types. Instead of only disease and chemical categories, future versions could extract symptoms, procedures, lab tests, body parts, vital signs, and social determinants of health. This would make the system more useful for real patient reports.

Third, long document handling should be improved. The current maximum sequence length may truncate long reports. A chunking strategy could split long reports into sections, run NER on each chunk, merge duplicate entities, and preserve the report structure.

Fourth, explanation quality should be improved. The fallback dictionary improves coverage, but it is manually curated and does not scale as well as retrieval from validated medical knowledge bases. A better system could combine MedQuAD, UMLS, MedlinePlus, RxNorm, and other trusted medical resources. Explanations could also include source labels and confidence scores.

Fifth, the interface could include uncertainty indicators. For example, each extracted entity could show model confidence, retrieval confidence, and whether the explanation came from MedQuAD or fallback. This would help users understand the reliability of each output.

Sixth, the system could include patient-friendly summarization. After extracting entities, a summarization module could generate a short explanation of the report. However, this should be done carefully to avoid unsupported medical claims.

Seventh, the system could be evaluated with real users. A usability study could measure whether users find the grouped output and explanations helpful. This would provide stronger evidence beyond model metrics.

VIII. RESPONSIBLE AI CONSIDERATIONS

Responsible AI is central to this project because the system operates in the biomedical domain. Even though MedAgent is not a diagnostic tool, users may still interpret medical explanations as advice. Therefore, the system must clearly communicate its intended use and limitations.

A. Non-Diagnostic Use

MedAgent is designed for entity extraction and explanation only. It should not be used to diagnose diseases, prescribe medication, recommend treatment, or replace a healthcare professional. The system can identify that a report mentions “asthma” or “hypertension,” but it cannot determine whether the patient currently has the condition or what treatment is appropriate.

B. Dataset Bias

The model is trained on BC5CDR, which is based on biomedical literature. Biomedical literature differs from real patient notes, discharge summaries, and patient-generated text. As a result, the model may perform better on formal biomedical phrasing than on informal or incomplete reports. The

dataset may also underrepresent certain conditions, populations, or language styles.

C. Explanation Reliability

MedQuAD explanations are useful, but retrieval errors can occur. A keyword-based retrieval system may return an explanation that is related to the entity but not perfectly matched to the report context. The fallback explanation layer improves coverage but introduces manually written content. Future versions should include source tracking and confidence estimates.

D. Privacy

The system is developed and evaluated using public and synthetic data. This reduces privacy risk. If the system were deployed with real patient reports, it would need strong privacy protections, secure storage, access control, and compliance with healthcare data regulations.

E. Transparency

The interface should clearly show what the system extracted and where explanations came from. A future version should label each explanation as “MedQuAD retrieved” or “fallback explanation.” It should also display uncertainty when no reliable explanation is found.

F. Human Oversight

Human oversight is necessary for biomedical AI systems. MedAgent should be used as a support tool for understanding text, not as an autonomous clinical system. Any medically significant interpretation should be reviewed by a qualified healthcare professional.

IX. CONCLUSION

This paper presented MedAgent, an end-to-end biomedical entity extraction and explanation system for patient reports. The system fine-tunes BioBERT on the BC5CDR dataset to identify disease and chemical entities. It then reconstructs entity spans, groups them into meaningful categories, retrieves explanations from MedQuAD, and displays the results through a Gradio interface.

The evaluation results show strong biomedical NER performance, with precision of 0.8416, recall of 0.8919, F1-score of 0.8660, and accuracy of 0.9665. Qualitative testing on asthma, heart disease, and general patient reports shows that the system can extract clinically meaningful terms and provide readable explanations. The addition of fallback explanations improves coverage for common medications and symptoms not reliably matched in MedQuAD.

This work demonstrates a complete AI lifecycle, including dataset preparation, model development, evaluation, deployment, interface design, and responsible AI reflection. Its main contribution is the integration of biomedical entity extraction and explanation retrieval into a practical, interactive system. Although the current prototype has limitations in semantic retrieval, entity coverage, and clinical reasoning, it provides a

strong foundation for future work in interpretable biomedical NLP systems.

Overall, MedAgent demonstrates that domain-specific language models can be combined with retrieval-based explanation resources to make biomedical text more structured and interpretable. The system does not replace clinical expertise, but it shows how automated entity extraction and explanation support can reduce the gap between complex medical terminology and user understanding. With improved semantic retrieval, expanded entity coverage, and stronger validation, the approach can be extended toward more robust biomedical report understanding systems.

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